AttriSelect: Efficient Attribute-Driven Data Selection for Targeted Vision Model Repair

Anonymous Author(s)
Affiliation
Address
email

Abstract

Targeted vision repair aims to correct model failures on rare semantic slices after a debugging pipeline identifies the failure slice and a generator produces corresponding repair images. In most existing workflows, these repair images are appended to the original training set, and the model is retrained or fine-tuned using all source data. While this preserves broad coverage, it also makes each repair cycle computationally expensive and time-consuming, especially as repairs accumulate over multiple iterations. Moreover, it provides little control over which source examples actually contribute to the repair. This naturally gives rise to a data selection problem related to active learning: rather than using the full dataset, the goal is to identify a compact yet informative subset that best supports repair while substantially reducing retraining cost. However, an effective repair subset must satisfy requirements beyond standard sample efficiency. In particular, it should preserve class coverage, retain rare repair-relevant attributes, and avoid redundancy with the generated repair data. Existing active learning methods typically prioritize uncertainty, diversity, or gradient-space coverage, but these signals are often poorly aligned with the semantic attributes that define the failure slice. To address this limitation, we propose AttriSelect, an attribute-driven selector that prioritizes samples based on attribute rarity, generated-data coverage, and class balance. We provide theoretical analysis and evaluate AttriSelect across image classification, pose estimation, and object detection. Across tasks and repair budgets, AttriSelect consistently achieves the strongest repair performance, demonstrating that explicit attribute metadata provides a repair-specific selection signal distinct from traditional active learning criteria.

1 Introduction

Computer vision models can achieve strong aggregate benchmark performance while still failing on rare semantic subpopulations [4, 6]. For example, a model may recognize people accurately overall but fail specifically on people wearing brown clothes in rainy scenes. Such failures are not fully characterized by the class label alone; instead, they arise from combinations of semantic attributes whose support in the training data may be limited [5, 9]. In deployed systems, the challenge is therefore not only to identify these failure slices, but also to repair them efficiently without repeatedly retraining on the entire dataset.

To better identify and characterize these failures, recent work has developed attribute-aware debugging methods that improve the interpretability of failure discovery. 3DB [15] constructs structured attribute spaces over model errors, while HiBug [6] leverages vision–language models to describe failure slices through shared visual attributes. Related approaches such as AdaVision [11], DCD [22], and HiBug2 [7] identify underperforming subpopulations and support retrieval-style interventions. In parallel, generation-based repair methods provide a complementary mechanism for correcting

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these failures. For example, HiBug [6] and subsequent methods [25, 23] synthesize targeted repair examples for identified slices, while SafeFix [23] further incorporates controlled image editing and vision–language filtering to preserve labels and unrelated attributes.

These pipelines address two parts of the repair workflow: identifying the failing semantic slice and generating targeted repair examples. However, a separate question remains: once targeted images are generated, how should they be combined with the original source data? Retraining on the full source pool can be costly, especially for dense prediction tasks. For example, training Keypoint R-CNN [13] for COCO person keypoint estimation [16] takes on the order of eight GPU-days [10]. As new bug slices are discovered across repair iterations, repeatedly retraining with the full source dataset becomes increasingly expensive.

Selecting a compact subset of source data is a natural strategy for reducing the cost of iterative repair, making the problem closely related to active learning (AL). Classical AL methods such as entropy-based uncertainty sampling [21], CoreSet [20], and BADGE [1] select compact training subsets using model uncertainty, feature-space coverage, or gradient embeddings. However, targeted repair introduces challenges beyond standard AL settings. The selector must preserve useful source coverage while allocating sufficient budget to the diagnosed attribute slice, and it must account for the distribution shift introduced by generated repair images. As a result, a generic uncertainty- or coverage-based score may select informative images while still omitting rare repair-relevant attribute values, whereas an attribute-greedy policy may over-select values already supplied by the generated data.

We study this remaining step as a repair-time source-selection problem. The input is not an unlabeled pool awaiting annotation, but a diagnosed failure slice, a generated repair set for that slice, and an annotated source pool from which only a compact subset can be retained. This changes the selection objective: the subset should preserve useful source coverage, include real examples with rare repair-relevant attributes, and avoid spending budget on attribute values already supplied by the generated images. We propose AttriSelect for this setting. Rather than representing source images only through feature embeddings or gradient statistics, AttriSelect treats each image as a structured attribute record and scores it using class-conditional rarity, generated-data frequency control, and soft class balancing. The method is therefore a focused selection component for existing repair pipelines: it does not replace failure-slice discovery or repair-image generation, but decides which original training examples should accompany the generated data during retraining.

Our contributions are:

1. We formulate targeted vision repair as a repair-time source-selection problem: given a diagnosed failure slice, generated repair data, and an annotated source pool, the goal is to select a compact source subset for retraining.
2. We propose AttriSelect, an attribute-driven source selector that combines class-conditional rarity, generated-data frequency control, and soft class balancing to prioritize repair-relevant source examples while avoiding redundancy and class collapse.
3. We provide a theoretical analysis of repair-slice enrichment, showing when attribute-aware selection preferentially allocates source budget to rare target slices, in contrast to attribute-agnostic selectors whose uncertainty or coverage signals are weakly aligned with the diagnosed failure slice.
4. We evaluate AttriSelect across image classification, pose estimation, and object detection, comparing against uncertainty-based, coverage-based, gradient-embedding, detector-oriented, and random or stratified baselines under matched repair budgets.

2 Related Work

Attribute-Based Debugging and Repair. Attribute-based debugging methods aim to expose model failures as human-interpretable semantic slices. AdaVision [11] and DCD [22] identify underperforming subpopulations and rely on retrieval-style interventions, which can require external data sources and manual validation. HiBug [6] uses vision–language models to discover failure slices through shared visual attributes, and HiBug2 [7] further improves the efficiency and coverage of this debugging process. SafeFix [23] shifts from retrieval to controlled generation by editing real training instances and filtering generated images with a vision-language model (VLM) to preserve

labels and unrelated attributes. These works focus primarily on finding bug slices or constructing repair examples. We use this line of work as the upstream repair setting and study the remaining selection problem: given original source data and targeted generated data, select a compact retraining subset that preserves useful coverage while emphasizing repair-relevant attributes.

Active Learning. Active learning reduces annotation or retraining cost by selecting informative subsets under a budget. Representative approaches include entropy-based uncertainty sampling [21], coreset selection [20], and hybrid gradient-embedding methods such as BADGE [1]. Recent object-detection and label-budget-robust variants combine these signals, for example PPAL for plug-and-play object detection [24] and Uncertainty Herding for uncertainty-weighted coverage across label budgets [2]. Our setting differs from standard AL because the goal is not generic sample efficiency, but selecting source data that complements a targeted repair set and corrects an attribute-defined failure slice.

3 Background

Attribute-Based Failure Slices. A failure slice is a group of examples that share a semantic description and have poor model performance. For instance, a pose model may fail on people with brown clothing in rainy scenes, even if its average performance is high. Attribute-based debugging methods make this description explicit by attaching semantic attributes to each image.

Formally, let $f_\theta : \mathcal{X} \to \mathcal{Y}$ be the vision model and let each image $x \in \mathcal{X}$ have attributes $\mathcal{A} = \{a_1, \ldots, a_m\}$. Each attribute $a_j : \mathcal{X} \to \mathcal{V}_j$ maps an image to a discrete value, such as a color, weather condition, texture, or viewpoint. A slice defined by several attribute values is $T = \{x \mid a_{j_1}(x) = v_{j_1}, \ldots, a_{j_k}(x) = v_{j_k}\}$. We focus on slices that are both rare in the training set $\mathcal{D}_{\text{train}}$ and weak on the validation set $\mathcal{D}_{\text{val}}$. In other words, the slice has low support and its validation accuracy is substantially below the overall validation accuracy.

Source Data and Generated Repair Data. After a slice has been diagnosed, repair uses two kinds of images. The first is the original source pool, denoted $\mathcal{D}_{\text{src}} \subseteq \mathcal{D}_{\text{train}}$. These images preserve the broad class and context coverage learned by the model. The second is a generated repair pool, denoted $\mathcal{D}_{\text{gen}}$. These images are produced for the diagnosed slice by pipelines such as HiBug and SafeFix [6, 23]; they are intended to satisfy the target attributes while preserving the original label.

Data Selection for Repair. The repair question is which source images should be kept when retraining with the generated images. Keeping all of $\mathcal{D}_{\text{src}}$ is expensive, but keeping an arbitrary small subset can remove useful coverage. We write the retraining set as $\mathcal{D}_{\text{repair}} = \mathcal{D}_{\text{sel}}^{\text{src}} \cup \mathcal{D}_{\text{sel}}^{\text{gen}}$, where $\mathcal{D}_{\text{sel}}^{\text{src}} \subseteq \mathcal{D}_{\text{src}}$ is the selected source subset and $\mathcal{D}_{\text{sel}}^{\text{gen}} \subseteq \mathcal{D}_{\text{gen}}$ is the generated subset used for repair. In our experiments, $\mathcal{D}_{\text{sel}}^{\text{gen}}$ is fixed to 1,000 generated images, so the main decision is the choice of $\mathcal{D}_{\text{sel}}^{\text{src}}$. A useful selector should keep class coverage, add real examples that complement the generated data, and avoid spending source budget on attribute values already abundant in $\mathcal{D}_{\text{gen}}$.

4 Methodology

4.1 AttriSelect Framework

AttriSelect is designed as the selection step inside an existing targeted repair pipeline. The upstream workflow can follow prior model-repair systems such as HiBug and SafeFix [6, 23]. Using the SafeFix-style ImageNet10 classification setting as a running example, a vision-language model (VLM) first proposes a task-level attribute set $\mathcal{A}$ and assigns an attribute value $v_{i,a}$ to each training or validation image. A ResNet classifier is then trained to a checkpoint θ_t, after which the repair pipeline analyzes the validation behavior of f_{θ_t}. An attribute slice T_{bug} is marked as a rare failure case if it is underrepresented in training and underperforms on validation:

$$|T_{\text{bug}} \cap \mathcal{D}_{\text{train}}| < \rho |\mathcal{D}_{\text{train}}|, \qquad \text{Acc}(T_{\text{bug}}) < \text{Acc}(\mathcal{D}_{\text{val}}), \tag{1}$$

where ρ is a user-specified rarity threshold. For example, the ImageNet10 pipeline may diagnose a slice whose attribute is *texture* and whose value is *fabric*. A conditional diffusion model, such as ControlNet [26], then generates repair images of objects with fabric texture while preserving the class label and other attributes.

AttriSelect starts after this upstream diagnosis and generation stage. It takes as input the diagnosed slice, the original source pool with VLM-assigned attributes, and the generated repair images for that slice. The output is a compact source subset $\mathcal{D}_{\text{sel}}^{\text{src}}$, which is combined with the generated images and used to continue training or fine-tune the checkpoint θ_t. Thus AttriSelect does not replace VLM-based attribute proposal, failure-slice discovery, or conditional image generation. It decides which original training examples should accompany the generated repair data so that retraining preserves source coverage while allocating more budget to real examples that complement the diagnosed repair slice.

4.2 Attribute-Driven Active Learning

Given $\mathcal{D}_{\text{src}}$ and $\mathcal{D}_{\text{gen}}$ for a discovered slice, the goal is to choose a small source subset that still supports the repair. Active learning is a natural tool for this problem: entropy-based uncertainty sampling selects examples with high predictive entropy [21], while CoreSet [20], BADGE [1], PPAL [24], and Uncertainty Herding [2] add coverage, gradient diversity, or detector-side difficulty signals.

For targeted repair, however, the subset must answer a semantic question: which real images best complement the generated examples for the diagnosed failure slice? A subset may provide good coverage of the source distribution in terms of class labels or feature space while still containing very few examples from the rare failure slice itself. Likewise, entropy-based sampling can miss the slice entirely if the model assigns higher uncertainty to unrelated images. Proposition 1 formalizes this base-rate mismatch.

A second challenge arises because generated repair images alter the attribute distribution seen during retraining. If the generated pool already contains many examples with a target attribute, the source selector should not continue allocating most of its remaining budget to redundant examples of the same attribute. For instance, if generated repair images already emphasize rainy weather, a purely attribute-greedy selector may over-select common rainy street scenes while failing to retain rainy examples from underrepresented classes, unusual viewpoints, or occluded settings. As a result, the repair process can become biased toward the easiest contexts in which the target attribute appears.

AttriSelect uses a three-part scoring function to avoid these failure modes. First, the **class-conditional rarity** term prioritizes attribute values that are rare within their class and therefore more likely to be repair-relevant. Second, the **generated-data frequency penalty** downweights attribute values that are already well represented by the generated repair images, reducing redundancy between synthetic and source data. Third, the **class-balance** term encourages coverage across classes by increasing the priority of classes that have received insufficient source budget. Together, these components steer selection toward repair-relevant attributes while preventing the subset from collapsing onto repeated synthetic attributes or a small number of easy classes.

Class-conditional rarity. For a source image x_i with class label y_i, let $v_{i,a}$ be the value of attribute a on that image. The rarity term gives higher weight to attribute values that occur fewer times within the same class:

$$R_{i,a} = \left(\frac{1}{n_a(v_{i,a} \mid y_i) + \varepsilon} \right)^{\alpha} \tag{2}$$

where $n_a(v_{i,a} \mid y_i)$ is the number of source examples in class y_i whose attribute a takes value $v_{i,a}$, $\varepsilon > 0$ is a numerical stability constant, and $\alpha \geq 0$ controls how strongly the selector prefers rare attribute values.

This term remains important even after adding generated repair images. Although generated images can make a target attribute less rare in the combined repair pool, they do not make real source examples with that attribute redundant, since the two data sources may differ in image statistics, contextual diversity, learned representations, and other unannotated visual factors. Because repair problems often originate from rare-case failures, class-conditional rarity helps retain real source examples containing rare repair-relevant attributes in the candidate ranking. The role of the generated-data frequency penalty is complementary: rather than promoting rarity, it prevents this preference from repeatedly selecting attribute values that are already heavily represented in the generated repair data.

Generated-data frequency penalty. Rarity alone can over-select attribute values that are already well covered by the generated repair pool. In the rainy-weather example above, if the generated data already contains many rainy images, the selector should not spend most of its remaining source budget

on additional easy rainy cases from similar contexts. Instead, it should retain useful real-source rainy examples while still leaving room for complementary classes, viewpoints, and occlusion patterns. To capture this trade-off, we measure how common each attribute value becomes after combining source and generated data as $\hat{n}_a(v) = n_a^{\mathrm{src}}(v) + \tilde{n}_a^{\mathrm{gen}}(v)$, where $n_a^{\mathrm{src}}(v)$ denotes the source count and $\tilde{n}_a^{\mathrm{gen}}(v)$ denotes the generated count used by the selector after any scaling of the generated pool. Attribute values with larger effective counts receive lower scores:

$$P_{i,a} = -\lambda \cdot \frac{\hat{n}_a(v_{i,a})}{\sum_{v \in \mathcal{V}_a} \hat{n}_a(v)} \tag{3}$$

where $\lambda \geq 0$ controls how strongly the selector penalizes already over-represented attribute values.

We then define the **base source score** $\bar{s}_i$ by summing the rarity and generated-data frequency terms over the repair attributes:

$$\bar{s}_i = \sum_{a \in \mathcal{A}} \Big(R_{i,a} + P_{i,a} \Big). \tag{4}$$

Each component addresses a different failure mode in repair-time selection. Setting $\alpha = 0$ removes the preference for rare semantic configurations within a class, reducing the selector to a frequency-driven policy. Setting $\lambda = 0$ removes the redundancy penalty, allowing the selected subset to repeatedly choose attribute values already well represented in $\mathcal{D}_{\mathrm{gen}}$. Without class-aware re-ranking, high-scoring examples may further concentrate within a small subset of classes. As a result, the score is not intended to produce a neutral representative subset; instead, it explicitly allocates source budget toward semantic coverage that is most useful for the diagnosed repair problem.

Class balance. Attribute scores are computed per image, so they can still over-select classes that happen to contain many high-scoring attributes. For example, in a fabric-texture repair, many high-scoring images may come from classes where fabric is visually common, while guitar images may contain the texture only in straps or carrying cases. We correct this with a soft class-balance term. Let n_{select} be the source-data budget and let N_{cls} be the number of classes, so the per-class target is $t = n_{\text{select}}/N_{\text{cls}}$. During greedy selection step k, a candidate from class c receives the adjusted score

$$\tilde{s}_i^{(k)} = \bar{s}_i + C_c^{(k)}, \qquad C_c^{(k)} = w \cdot \frac{t - n_c^{(k)}}{t}, \tag{5}$$

where $n_c^{(k)}$ is the number of already selected source samples from class c, and $w \geq 0$ controls the strength of class balancing. Classes below their target allocation receive a positive correction, while classes above their target receive a negative one. As w increases, the selector behaves more like stratified sampling, with the limiting case $w = \infty$ corresponding to round-robin class quotas. This mechanism preserves the attribute-based ranking from Eq. 4 within each class while preventing a small subset of classes from dominating the repair set.

Selection is performed greedily. We first compute the base score $\bar{s}_i$ for every source candidate and initialize both the selected set and the per-class counts to zero. At each iteration, every remaining candidate receives the adjusted score $\tilde{s}_i^{(k)}$ from Eq. 5. The highest-scoring candidate is then added to $\mathcal{D}_{\mathrm{sel}}^{\mathrm{src}}$, its class count is updated, and the class-balance corrections are recomputed. The process repeats until n_{select} source examples have been selected. The generated repair set remains fixed throughout this procedure and influences selection only through the frequency counts in Eq. 3. As a result, all methods are evaluated under the same generated-data budget.

4.3 Theoretical Analysis

In targeted repair, useful subsets should contain rare, underperforming semantic configurations that complement the generated data. Let $N = |\mathcal{D}_{\mathrm{src}}|$ be the source-pool size and let the source-selection budget be $m = \lfloor rN \rfloor$ for $0 < r < 1$. Let $T \subseteq \mathcal{D}_{\mathrm{src}}$ denote the target failure slice, and write $T_c = T \cap \{x_i : y_i = c\}$ and $N_c = |\{x_i : y_i = c\}|$ for its class-conditional support.

Proposition 1 (Attribute-agnostic AL requires target alignment)**.** *Consider any AL selector that, within class c, chooses B_c source examples using only non-attribute information h_i, such as model uncertainty, logits, feature embeddings, gradient embeddings, losses, detector-side difficulty scores, proxy class-quality weights, coverage gains, or image-level distances, and does not use the attribute values $\{v_{i,a} : a \in \mathcal{A}\}$. Let $\mathcal{D}_{\mathrm{AL}}$ denote its selected source subset. If the target-slice indicator*

$\mathbf{1}[x_i \in T_c]$ *is conditionally independent of the selector information* h_i *given class* c*, with ties and any internal randomness independent of the target indicator, then*

$$\mathbb{E}\big[|\mathcal{D}_{\mathrm{AL}} \cap T_c|\big] = B_c \frac{|T_c|}{N_c}. \tag{6}$$

Thus an attribute-agnostic AL method can enrich the target slice only when its non-attribute signal is additionally aligned with that slice.

The expectation in Eq. 6 reflects a base-rate limitation. For instance, if the target slice constitutes only 2% of class c and an AL method selects 100 examples from that class, the expected number of target-slice examples is just 2. This highlights the key advantage of AttriSelect in the repair setting: AttriSelect directly observes the semantic attributes defining the diagnosed slice, allowing selection to explicitly prioritize rare repair-relevant values rather than depending on uncertainty, difficulty, or coverage signals to correlate with them indirectly. The generated-data penalty and class-balance term further ensure that this preference does not degenerate into repeatedly selecting synthetic-like attributes or collapsing onto a small subset of classes. Coverage-based selectors face a related problem even when the target examples are present in the source pool. Suppose there are a few fabric-texture images, but their feature embeddings occupy a compact region. Traditional active learning methods may judge one or two representatives sufficient to cover that region. For generic active learning, this can be a reasonable allocation. For repair, however, the compact region is exactly the semantic slice where the model fails, so the retraining set may need many more fabric-texture examples than feature-space coverage alone would select.

Proposition 1 is formulated at the level of selector information rather than for a specific baseline method. Stronger baselines such as PPAL and Uncertainty Herding improve upon plain uncertainty sampling or coreset selection, but they still rely on uncertainty, difficulty, or coverage signals being aligned with the discovered attribute slice. When the failure slice is defined by rare semantic attributes that are weakly reflected in these signals, the expected allocation of selected samples to the target slice remains proportional to the slice prevalence. Appendix B provides the proof, a coverage-based instance, and the corresponding guarantees for the AttriSelect score. In particular, under explicit conditions on attribute rarity and generated-data frequency, AttriSelect ranks rare target-slice examples above non-target examples within the same class, enriches the selected repair subset with repair-relevant attributes, and controls class imbalance through the soft class-balance term.

5 Experiments

Datasets and Tasks We evaluate AttriSelect on three vision tasks: image classification, pose estimation, and object detection. In all settings, the goal is to test whether attribute-aware source-data selection can choose a subset that improves the model performance and fixes bugs.

For image classification, we follow SafeFix [23]: a ResNet-18 classifier on an ImageNet subset [8] with 10 classes, 1,300 images per class, and an 8:1:1 train/validation/test split, denoted ImageNet10. We also evaluate CelebA [17] lipstick-wearing classification with a ResNet-18-based binary classifier and 80,000/10,000/10,000 train/validation/test images. For pose estimation, we follow HiBug2 [7] on COCO person keypoints [16], using Keypoint R-CNN [13] with a ResNet-50-FPN backbone, 20,000 source training images, and 1,173-image held-out validation and test splits, evaluated by COCO keypoint AP. For object detection, we use KITTI [12] Car and Pedestrian instances converted to YOLO format with 5,000/1,240/1,241 train/validation/test images, and train YOLOv8n.

Metrics Let M denote the task metric: test accuracy for image classification, COCO keypoint AP for pose estimation, and mAP50–95 for object detection. We focus on targeted improvements for failure cases, which are critical for fairness and safety, while also tracking overall task performance. Following prior work [14, 23], we report the Fix Rate (FR),

$$\mathrm{FR}(\%) = 100 \cdot \frac{M_{\mathrm{after}} - M_{\mathrm{before}}}{1 - M_{\mathrm{before}}}, \tag{7}$$

where M_{before} and M_{after} are the test metrics before and after repair. For accuracy, FR is the percentage of the original error gap that is corrected; for AP and mAP, it is the normalized improvement toward the maximum score, reported in percent.

Table 1: ImageNet10 image-classification repair results. Metric: FR (%).

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	**Ours**
0.1	3.40	3.47	2.87	3.40	5.31	5.59	3.96	**6.93**
0.3	4.78	6.93	4.21	6.93	8.84	10.47	7.75	**11.28**
0.5	7.32	9.66	5.59	8.03	10.47	8.38	8.84	**11.85**
0.7	8.84	6.12	6.40	7.75	10.75	10.47	8.84	**12.10**
0.9	10.22	10.47	10.22	6.12	9.37	11.57	11.85	**13.76**
1.0	8.03	8.03	8.03	8.03	8.03	8.03	8.03	8.03

Table 2: CelebA image-classification repair results. Metric: FR (%).

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	**Ours**
0.1	4.88	2.44	7.10	-6.04	3.18	-1.06	5.94	**8.48**
0.3	6.04	5.30	8.70	7.95	1.91	3.50	4.03	**8.91**
0.5	6.04	7.95	7.64	8.17	2.86	2.23	9.97	**13.79**
0.7	2.86	8.27	10.18	6.57	8.06	5.09	6.04	**12.09**
0.9	3.18	4.56	8.27	6.04	6.04	4.56	6.57	**14.74**
1.0	6.04	6.04	6.04	6.04	6.04	6.04	6.04	6.04

Baselines We compare AttriSelect with seven generic source-selection baselines. RANDOM samples source images uniformly at random. STRATIFIED preserves class-level proportions during sampling [18]. ENTROPY is the entropy-based uncertainty baseline; it selects examples with the largest predictive entropy or task-specific confidence-entropy proxy [21]. CORESET uses greedy coverage in embedding space [20]. BADGE selects gradient-embedding batches that combine uncertainty and diversity [1]. PPAL is a plug-and-play active-learning baseline for object detection [24]. UHERDING denotes Uncertainty Herding, which combines uncertainty with coverage across label budgets [2].

5.1 Main Results

Tables 1–4 compare AttriSelect with generic source-selection baselines using Fix Rate. The ratio 1.0 rows use the full original source set, so subset selection is inactive and the repeated entries are the corresponding full-data reference. The original task metrics are reported in Appendix C.

AttriSelect gives the best FR in 19 of the 20 active source-selection settings, excluding the ratio-1.0 full-data references where all selectors are identical. The improvement is clear at small budgets, where the selected source subset is highly constrained, and it generally remains at larger budgets. KITTI is the only near-exception: PPAL is slightly higher at ratio 0.7 by 0.05 FR points, while AttriSelect remains best at the other four KITTI ratios.

Subset selection can also outperform the full-data setting, especially on ImageNet10, CelebA, and COCO. This suggests that the full candidate pool contains noisy, redundant, or target-irrelevant samples, and that selection improves the effective training distribution. In this sense, AttriSelect is not only a way to save data or training cost; it also improves data quality and target-slice alignment during repair.

5.2 Computational Efficiency

AttriSelect is also cheaper than generic active-learning selectors. Its selection cost is dominated by counting VLM-assigned attribute values and sorting candidates, while uncertainty, coverage, and gradient-embedding baselines require model scoring, embedding comparisons, or iterative diversity updates. Table 5 reports source-subset selection time on COCO and CelebA for the three ratios used in the main small-to-medium budget comparison. The source pools contain 20,000 COCO training images and 80,000 CelebA training images. Random and stratified sampling have negligible algorithmic selection cost after excluding data loading and export overhead; we keep them in the table as zero-cost references.

Table 3: COCO pose-estimation repair results. Metric: FR (%).

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	**Ours**
0.1	3.56	5.08	5.61	-0.61	2.03	2.71	1.17	**6.86**
0.3	3.37	8.32	7.64	8.64	6.12	4.32	6.42	**9.86**
0.5	7.49	8.00	9.27	6.47	7.42	6.12	6.83	**10.13**
0.7	6.08	9.69	9.54	9.10	8.64	6.17	7.83	**10.18**
0.9	4.61	9.49	6.54	7.32	7.59	7.05	9.74	**11.05**
1.0	8.71	8.71	8.71	8.71	8.71	8.71	8.71	8.71

Table 4: KITTI object-detection repair results. Metric: FR (%).

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	**Ours**
0.1	-9.65	-7.27	-7.52	-8.89	-6.95	-9.24	-8.32	**-5.81**
0.3	-0.48	-1.23	0.23	-0.32	-0.05	-1.21	-1.33	**0.46**
0.5	2.13	1.23	3.20	3.04	2.29	1.12	2.17	**3.36**
0.7	3.20	3.06	3.34	3.34	2.08	3.25	**3.57**	3.52
0.9	4.37	4.66	4.66	3.54	4.21	4.85	3.89	**5.12**
1.0	5.30	5.30	5.30	5.30	5.30	5.30	5.30	5.30

Across these settings, AttriSelect selects the source subset in less than two seconds, whereas the fastest non-attribute active-learning baseline takes tens of seconds and the slower coverage or gradient-embedding methods can take minutes to hours. The efficiency gain is therefore not only relative to the most expensive baselines: even against the fastest nontrivial baseline in each row, AttriSelect is at least $36\times$ faster on CelebA and more than $230\times$ faster on COCO.

5.3 Attribute-Slice Accuracy Deltas

Figure 1 reports attribute-slice accuracy deltas, defined as the accuracy of the retrained model minus the accuracy of the original ResNet checkpoint θ_t on the same slice. All methods use the same generated repair images and differ only in the selected ratio-0.5 source subset. The generated data targets pink color and fabric texture. AttriSelect gives the largest gains on the targeted attribute families, especially the exact repair values pink and fabric, indicating that the selected source subset improves target-slice alignment rather than only changing overall accuracy. A COCO pose-estimation version with the full annotated attribute vocabulary is provided in Appendix D.

5.4 Ablation Study

We ablate the three score components on ImageNet10 ControlNet repair to test whether the gains come from a single term or from their combination. Table 6 reports test accuracy for all component subsets. The full $R + P + C$ setting combines rarity R, generated-data frequency penalty P, and class balance C.

The full $R + P + C$ setting is the most reliable pattern, giving the best result at every ratio. The leave-one-component-out drops from the full setting are nearly identical: removing R drops test accuracy by 0.00942 on average, removing P by 0.00940, and removing C by 0.00940. Thus the ablation supports a complementarity claim rather than a single dominant-component claim on ImageNet10. Single-component variants are weaker on average (R: 0.74198, P: 0.74038, C: 0.73814) than the full setting (0.74892), showing that each component alone is insufficient. A COCO Brown-Rainy component ablation with a broadly similar trend is reported in Appendix F.

6 Conclusion

We studied targeted vision model repair as a repair-time source-selection problem. In this setting, a failure slice has already been diagnosed and targeted repair images have already been generated; the remaining question is which original source examples should accompany those generated images during retraining. AttriSelect addresses this question with an attribute-driven score that combines class-conditional rarity, generated-data frequency control, and soft class balancing. The analysis

Table 5: Selection-stage wall-clock time. Units are seconds (s), minutes (m), and hours (h). Speedup is measured against the fastest non-attribute active-learning baseline in the same row.

Dataset	Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	**Ours / speedup**
COCO	0.1	0s	0s	863s	5m	21m	86s	67s	**0.29s / 231**×
COCO	0.3	0s	0s	865s	15m	1h	93s	85s	**0.30s / 283**×
COCO	0.5	0s	0s	865s	25m	2h	94s	90s	**0.32s / 281**×
CelebA	0.1	0s	0s	49s	45m	2h	63s	46s	**0.42s / 110**×
CelebA	0.3	0s	0s	51s	3h	5h	65s	51s	**0.51s / 100**×
CelebA	0.5	0s	0s	49s	5h	9h	61s	61s	**1.37s / 36**×

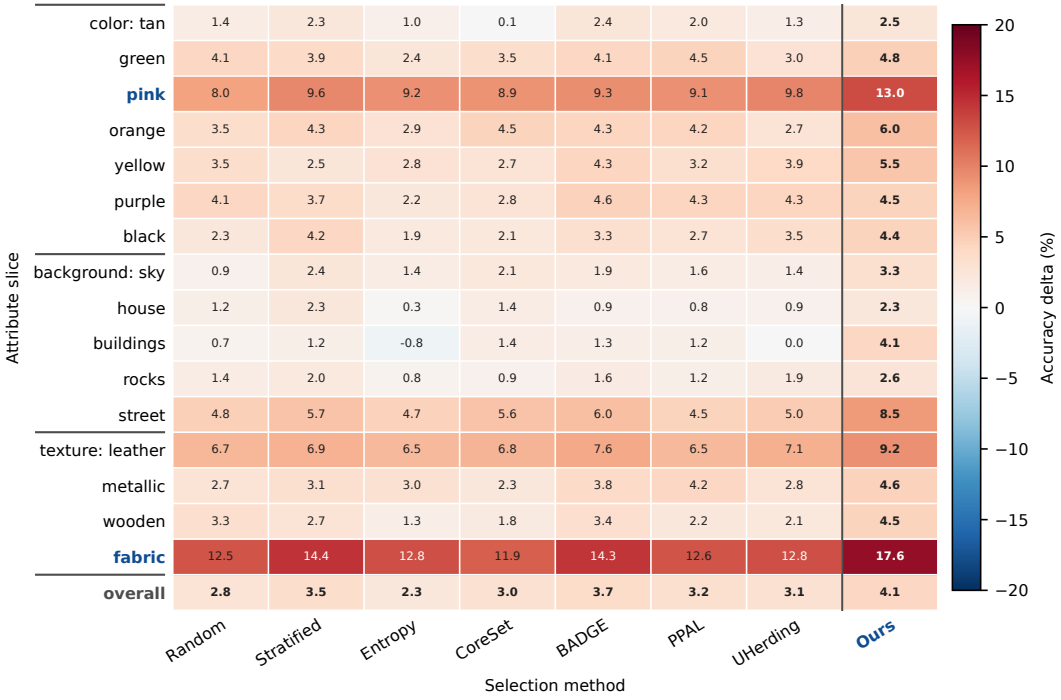


Figure 1: ImageNet10 ratio-0.5 attribute-slice accuracy deltas. The upstream repair pipeline generates examples targeting pink color and fabric texture, and each method retrains with the same generated data plus a ratio-0.5 selected source subset. AttriSelect gives larger gains on the targeted attribute families, with the strongest gains on the exact repair values.

shows when this score enriches rare target slices relative to attribute-agnostic selectors, and the experiments across classification, pose estimation, and detection indicate that structured attribute metadata provides a useful signal beyond generic uncertainty, coverage, and gradient-embedding scores. This makes AttriSelect a focused component for localized repair pipelines rather than a replacement for upstream bug discovery or image generation.

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Table 6: ImageNet10 ControlNet attribute-score component ablation. Metric: test accuracy.

Ratio	none	R	P	C	$P+C$	$R+C$	$R+P$	$R+P+C$
0.1	0.7269	0.7300	0.7262	0.7277	0.7359	0.7308	0.7315	**0.7369**
0.3	0.7308	0.7415	0.7392	0.7392	0.7308	0.7369	0.7315	**0.7492**
0.5	0.7400	0.7415	0.7408	0.7346	0.7369	0.7415	0.7446	**0.7508**
0.7	0.7423	0.7469	0.7431	0.7415	0.7431	0.7469	0.7477	**0.7515**
0.9	0.7462	0.7500	0.7526	0.7477	0.7508	0.7415	0.7423	**0.7562**

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Appendix

A Implementation Details

We use vision-language models (VLMs) in two separate roles. First, a VLM proposes the semantic attribute vocabulary $\mathcal{A}$ for a task. For example, in KITTI object detection, GPT-5 [19] is used to define candidate visual attributes such as color, weather, scale, viewpoint, occlusion, truncation, and motion state. Second, a VLM assigns attribute values to the source images. In our KITTI implementation, we use Qwen3-VL [3] for this assignment step. For the red-cloudy KITTI repair, color and weather are the attribute fields tied to the discovered repair slice.

B Additional Theory Analysis

Slice enrichment We use *slice enrichment* to describe whether a selected source subset contains more examples from the diagnosed target slice than would be expected from its class-conditional base rate.

Proof of Proposition 1 Conditional independence makes the target indicators exchangeable with respect to any ranking, batch construction, or randomized choice computed from h_i inside class c. The B_c selected positions therefore contain, in expectation, the same target-slice fraction as the class pool itself, namely $|T_c|/N_c$. This argument does not depend on the particular AL heuristic. Uncertainty sampling, margin sampling, loss-based selection, diversity sampling, CoreSet, BADGE, PPAL-style uncertainty-then-diversity selection, and Uncertainty Herding-style uncertainty coverage all fall under the claim whenever their selection variables omit the current attribute metadata and are not otherwise aligned with the target slice.

As a concrete coverage-based instance, let z_i be the representation used by a diversity or coverage selector, such as cluster-based Diversity, CoreSet in feature space, or BADGE in gradient-embedding space. CoreSet-style k-center selection minimizes

$$\max_i \min_{j \in S} d(z_i, z_j), \tag{8}$$

so achieving objective value at most ϵ is equivalent to constructing an ϵ-cover of the source representations. For a finite set A in this representation space, define the empirical covering number

$$\mathcal{N}_\epsilon^{\mathrm{emp}}(A) = \min\big\{|C| : C \subseteq A,\ A \subseteq \cup_{z \in C} B(z, \epsilon)\big\}. \tag{9}$$

If the target slice in class c has empirical covering number

$$q_c = \mathcal{N}_\epsilon^{\mathrm{emp}}\big(\{z_i : x_i \in T_c\}\big), \tag{10}$$

then an ϵ-cover objective can cover the entire target region using only q_c selected target representatives. Consequently, when the non-target region still contains uncovered ϵ-scale cells, a coverage-based selector has an optimal solution with at most q_c selected target examples, even if $|T_c| \gg q_c$.

For this coverage-based instance, take $h_i = z_i$. By definition of the empirical covering number, there exists a subset $C_T \subseteq \{z_i : x_i \in T_c\}$ with $|C_T| = q_c$ such that every target-slice representation is within distance ϵ of some point in C_T. Selecting the corresponding q_c target examples therefore makes the whole target slice ϵ-covered. Any additional target example only refines coverage inside an already covered target region. If non-target ϵ-scale cells remain uncovered, a coverage or diversity objective can spend the remaining representatives outside T_c without increasing its maximum covering radius. Thus feature-space coverage does not force more than q_c target selections.

The same conclusion follows from the usual volumetric intuition. If a class region has effective dimension d, volume V, and the target subregion has volume V_T, then away from boundary effects an ϵ-cover requires on the order of $V/(v_d\epsilon^d)$ cells for the full class region but only $V_T/(v_d\epsilon^d)$ cells for the target region, where v_d is the volume of the unit d-ball. When the bug slice is semantically important but geometrically compact, $V_T \ll V$, so the number of representatives allocated by a geometry-only objective can be small. The objective allocates capacity to representation volume, not to semantic repair importance.

Rarity and generated-data penalty We now show why the attribute score assigns strict priority to a rare target slice under explicit data conditions. For any target example $x_i \in T_c$ and non-target example $x_j \notin T_c$ from the same class c, define the attributes on which they differ as

$$\mathcal{D}_{ij} = \{a \in \mathcal{A} : v_{i,a} \neq v_{j,a}\}. \tag{11}$$

Let

$$A_{ij} = \sum_{a \in \mathcal{D}_{ij}} \left[\left(\frac{1}{n_a(v_{i,a} \mid c) + \varepsilon} \right)^\alpha - \left(\frac{1}{n_a(v_{j,a} \mid c) + \varepsilon} \right)^\alpha \right] \tag{12}$$

be the rarity advantage of the target example, and let

$$F_{ij} = \sum_{a \in \mathcal{D}_{ij}} \left[\frac{\hat{n}_a(v_{i,a})}{\sum_{v \in \mathcal{V}_a} \hat{n}_a(v)} - \frac{\hat{n}_a(v_{j,a})}{\sum_{v \in \mathcal{V}_a} \hat{n}_a(v)} \right] \tag{13}$$

be the frequency-penalty disadvantage introduced by source and generated counts. If generated data has flooded the target value, then F_{ij} can be positive; if not, the penalty does not hurt the target example relative to x_j.

Proposition 2 (Rarity-dominates-penalty margin)**.** *Assume $\alpha > 0$ and $A_{ij} > 0$ for every target/non-target pair (x_i, x_j) within class c. Define*

$$\lambda_c^\star = \min_{x_i \in T_c,\ x_j \notin T_c,\ F_{ij} > 0} \frac{A_{ij}}{F_{ij}}, \tag{14}$$

with $\lambda_c^\star = \infty$ if no pair has $F_{ij} > 0$. For any $0 \leq \lambda < \lambda_c^\star$, every target example in T_c has larger base score $\bar{s}_i$ than every non-target example from the same class. In particular, when each target attribute value is rarer than the competing non-target value, i.e., $n_a(v_{i,a} \mid c) < n_a(v_{j,a} \mid c)$ for all $a \in \mathcal{D}_{ij}$, the condition $A_{ij} > 0$ holds automatically.

Proof. For a target/non-target pair (x_i, x_j) in class c, attributes outside $\mathcal{D}_{ij}$ contribute the same score to both examples and cancel. The remaining base-score gap is

$$\bar{s}_i - \bar{s}_j = A_{ij} - \lambda F_{ij}. \tag{15}$$

If $F_{ij} \leq 0$, then $\bar{s}_i - \bar{s}_j \geq A_{ij} > 0$ for any $\lambda \geq 0$, so the frequency penalty does not overturn the rarity advantage. If $F_{ij} > 0$, then $\bar{s}_i - \bar{s}_j > 0$ whenever $\lambda < A_{ij}/F_{ij}$. Taking the minimum over all target/non-target pairs with $F_{ij} > 0$ gives the interval $0 \leq \lambda < \lambda_c^\star$, under which every pair has a strictly positive score gap. Finally, if $n_a(v_{i,a} \mid c) < n_a(v_{j,a} \mid c)$ and $\alpha > 0$, then each rarity term in A_{ij} is positive, so their sum is positive. □

This proposition is the main data-dependent guarantee. It says that if the VLM annotations correctly identify a genuinely rare target slice, then there is a nonempty range of λ values for which rarity dominates the synthetic-data frequency penalty. Even when generated examples make the target value more frequent in the repair candidate pool, the target examples remain ranked above non-target examples as long as λ is below the explicit threshold $\lambda_c^\star$. Conversely, if target and non-target samples are indistinguishable under $\mathcal{A}$, then no attribute-only selector can separate them without additional information.

Proposition 3 (Deterministic rare-slice enrichment)**.** *Assume the rarity-dominates-penalty condition in Proposition 2 holds for class c and the λ used by the selector. If the class-aware re-ranking step allocates B_c source samples to class c, then AttriSelect includes at least*

$$\min\{B_c, |T_c|\} \tag{16}$$

target-slice examples from that class. A random or stratified selector with the same class allocation selects

$$\mathbb{E}\big[|S_{\text{rand}} \cap T_c|\big] = B_c \frac{|T_c|}{N_c} \tag{17}$$

target examples in expectation.

Proof. At a fixed greedy step, the class-aware term adds the same bonus $C_c^{(k)}$ to every remaining candidate from class c. Therefore it can change how many samples are selected from each class, but it cannot reverse the within-class ordering induced by the base attribute score $\bar{s}_i$. By Proposition 2, all target-slice examples in class c appear before all non-target examples in the class queue. Selecting B_c samples from that queue must therefore include all target examples when $|T_c| \leq B_c$, and B_c target examples when $|T_c| > B_c$. This proves the deterministic bound. For random or stratified sampling under the same class allocation, each selected class-c source example has probability $|T_c|/N_c$ of belonging to the target slice, giving the expectation by linearity. □

The enrichment gap is strict whenever $0 < |T_c| < N_c$ and $B_c > 0$, because

$$\min\{B_c, |T_c|\} \geq B_c \frac{|T_c|}{N_c}. \tag{18}$$

Thus AttriSelect provides a semantic coverage guarantee for the repair surrogate, while random, stratified, uncertainty, Diversity, CoreSet, BADGE, PPAL, and Uncertainty Herding require additional alignment between their selection signal and the target slice.

Proposition 4 (Uncertainty can miss an attribute slice)**.** *Consider uncertainty-based selection within each class, with uncertainty score Q_i and class allocation B_c. If the B_c largest uncertainty scores among non-target examples in class c are all larger than every uncertainty score in T_c, then uncertainty selection chooses zero examples from T_c.*

Proof. The claim is deterministic. If at least B_c non-target examples have uncertainty larger than every target example, the top-B_c uncertainty set contains only non-target examples. The same argument applies to any two-stage uncertainty-first method when the first-stage pool contains no target-slice examples: no second-stage diversity or coverage rule can recover target support that is absent from its candidate pool. □

Finally, suppose the repair utility of a selected source subset is

$$U(S) = u_0|S| + (u_T - u_0)|S \cap T|, \qquad u_T > u_0, \tag{19}$$

meaning target-slice samples have larger marginal value for the discovered repair problem. Then the rare-slice enrichment bound directly implies expected utility at least as large as random or stratified selection under the same class allocation, with a strict gap whenever the enrichment gap is strict. This is not a universal proof that the final trained model must dominate every baseline for every ratio; it is a conditional guarantee that the selector spends a fixed budget on more repair-relevant examples when the VLM metadata and count conditions correctly identify the failing slice.

Class-balance behavior The enrichment result above is within-class: the class-balance term changes how many examples are drawn from each class, but it does not reverse the attribute ordering inside a class. The following two results summarize the role of C.

Proposition 5 (Hard-balance limit)**.** *When $w = \infty$ and every active class has enough candidates, the round-robin limit selects class counts that differ by at most one. Within each class, candidates are still ordered by the attribute score $\bar{s}_i$.*

Proof. The round-robin rule visits one active class at a time and removes the highest-scoring remaining candidate from that class. After each visit, the class is moved to the back of the queue. Thus no class can receive a second additional sample before every other non-empty class has received one. The selected counts across non-empty classes therefore differ by at most one, except when a class runs out of candidates. Since each class queue is sorted by $\bar{s}_i$, the within-class attribute order is preserved. □

Proposition 6 (Finite-weight balance pressure)**.** *Assume $0 < w < \infty$. At greedy step k, let $A_c^{(k)}$ be the highest remaining base score in class c, and assume all active class-head scores lie in an interval of width*

$$\Delta = \max_c A_c^{(k)} - \min_c A_c^{(k)}. \tag{20}$$

If class d is selected at step k while another class c is also active, then

$$n_d^{(k)} - n_c^{(k)} \le \frac{t\Delta}{w}. \tag{21}$$

Equivalently, if an active class c trails class d by more than $t\Delta/w$, then class d cannot be selected before class c at that step.

Proof. Since class d is selected, its adjusted score is at least the adjusted score of class c:

$$A_d^{(k)} + w\frac{t - n_d^{(k)}}{t} \ge A_c^{(k)} + w\frac{t - n_c^{(k)}}{t}. \tag{22}$$

Rearranging gives

$$\frac{w}{t}\big(n_d^{(k)} - n_c^{(k)}\big) \le A_d^{(k)} - A_c^{(k)}. \tag{23}$$

Because both class-head scores lie in an interval of width Δ, we have $A_d^{(k)} - A_c^{(k)} \le \Delta$. Therefore $n_d^{(k)} - n_c^{(k)} \le t\Delta/w$. □

This finite-weight result gives the exact sense in which w controls soft balance. When $w = 0$, the bound disappears and the selector reduces to pure attribute ranking. When $0 < w < \infty$, a class with a large deficit receives enough bonus to overcome any bounded attribute-score advantage from an over-selected class. Increasing w tightens the admissible count gap by the factor $1/w$. In the limiting case $w \to \infty$, the admissible gap goes to zero, matching the round-robin behavior in Proposition 5. Thus finite w does not guarantee exact class quotas, but it guarantees that class imbalance can persist only when the over-selected class has a sufficiently larger attribute score at the current step.

B.1 No-Generation Corner Case

This section discusses the corner case in which no generated repair images are available, i.e., $\mathcal{D}_{\text{gen}} = \emptyset$. This setting removes the main mechanism for adding missing semantic support, but AttriSelect still has a useful source-selection interpretation. Since $\tilde{n}_a^{\text{gen}}(v) = 0$, the effective count becomes

$$\hat{n}_a(v) = n_a^{\text{src}}(v), \tag{24}$$

and the base score reduces to

$$\bar{s}_i = \sum_{a \in \mathcal{A}} \left[\left(\frac{1}{n_a(v_{i,a} \mid y_i) + \varepsilon} \right)^{\alpha} - \lambda \frac{n_a^{\text{src}}(v_{i,a})}{\sum_{v \in \mathcal{V}_a} n_a^{\text{src}}(v)} \right]. \tag{25}$$

Thus, when $\mathcal{D}_{\text{gen}} = \emptyset$, the frequency penalty is no longer correcting a synthetic flood. It becomes a source-distribution head-value penalty: values that are already common in the source pool receive lower priority. The rarity term still prefers values that are rare within a class, and the class-balance term still controls label-level collapse.

Proposition 7 (Source-only rare-slice enrichment)**.** *Let $\mathcal{D}_{\text{gen}} = \emptyset$ and let $T \subseteq \mathcal{D}_{\text{src}}$ be a target failure slice with nonzero source support. Suppose the rarity-dominates-penalty condition in Proposition 2 holds for T under the source-only score above, with $\hat{n}_a(v) = n_a^{\text{src}}(v)$. If the class-aware re-ranking step selects B_c samples from class c, then AttriSelect returns at least*

$$\sum_c \min\{B_c, |T_c|\} \tag{26}$$

source examples from T. A random selector with the same per-class allocation selects

$$\mathbb{E}\big[|S_{\text{rand}} \cap T|\big] = \sum_c B_c \frac{|T_c|}{N_c} \tag{27}$$

target-slice examples in expectation.

Proof. The proof is identical to the rare-slice enrichment proof in Appendix B, because that argument uses only the within-class ordering induced by $\bar{s}_i$. When $\mathcal{D}_{\text{gen}} = \emptyset$, the score changes only through the replacement $\hat{n}_a(v) = n_a^{\text{src}}(v)$. If the source-only score satisfies the rarity-dominates-penalty condition, then all target-slice samples appear before non-target samples in each class queue. Selecting B_c examples from class c therefore includes $\min\{B_c, |T_c|\}$ target examples. The random expectation follows by linearity under sampling from each class with allocation B_c. □

This proposition shows that generation is not required for the selector to improve target-slice coverage when the source pool already contains target examples. In this regime, the method becomes semantic active subset selection over the source pool. It can increase the number of existing rare failure cases in the retraining subset, but it cannot create support for a target slice that is absent from the source pool.

Proposition 8 (Impossibility for absent source slices)**.** *Let $\mathcal{D}_{\text{gen}} = \emptyset$ and suppose $T \cap \mathcal{D}_{\text{src}} = \emptyset$. Then every source-only selector $S \subseteq \mathcal{D}_{\text{src}}$ satisfies*

$$|S \cap T| = 0. \tag{28}$$

Consequently, no active-selection rule over $\mathcal{D}_{\text{src}}$ alone can provide target-slice training support.

Proof. Since $S \subseteq \mathcal{D}_{\text{src}}$ and $T \cap \mathcal{D}_{\text{src}} = \emptyset$, we have $S \cap T \subseteq \mathcal{D}_{\text{src}} \cap T = \emptyset$. Hence $|S \cap T| = 0$ for every possible selector. □

The no-generation case therefore separates two roles of AttriSelect. If T already appears in the source pool, attribute selection can enrich those existing target examples under the same rarity-dominates-penalty condition as the main result. If T is absent from the source pool, selection alone is insufficient; repair requires either new data collection or generated images that add target-slice support. This also clarifies the role of λ: with no generated images, a large frequency penalty may suppress globally common values even when they are rare within a specific class. In source-only repair, λ should therefore be interpreted as a head-value suppression weight and may need to be smaller than in settings with heavy generated-data injection.

C Main-Result Metric Tables

This section reports the original task metrics corresponding to the Fix Rate tables in the main text. The metrics are test accuracy for ImageNet10 and CelebA, COCO keypoint AP for pose estimation, and test mAP50–95 for KITTI object detection.

Table 7: ImageNet10 ControlNet repair results. Metric: test accuracy.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	**Ours**
0.1	0.7269	0.7271	0.7254	0.7269	0.7323	0.7331	0.7285	**0.7369**
0.3	0.7308	0.7369	0.7292	0.7369	0.7423	0.7469	0.7392	**0.7492**
0.5	0.7380	0.7446	0.7331	0.7400	0.7469	0.7410	0.7423	**0.7508**
0.7	0.7423	0.7346	0.7354	0.7392	0.7477	0.7469	0.7423	**0.7515**
0.9	0.7462	0.7469	0.7462	0.7346	0.7438	0.7500	0.7508	**0.7562**
1.0	0.7400	0.7400	0.7400	0.7400	0.7400	0.7400	0.7400	0.7400

Table 8: CelebA lipstick-wearing classification repair results. Metric: test accuracy.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	**Ours**
0.1	0.9103	0.9080	0.9124	0.9000	0.9087	0.9047	0.9113	**0.9137**
0.3	0.9114	0.9107	0.9139	0.9132	0.9075	0.9090	0.9095	**0.9141**
0.5	0.9114	0.9132	0.9129	0.9134	0.9084	0.9078	0.9151	**0.9187**
0.7	0.9084	0.9135	0.9153	0.9119	0.9133	0.9105	0.9114	**0.9171**
0.9	0.9087	0.9100	0.9135	0.9114	0.9114	0.9100	0.9119	**0.9196**
1.0	0.9114	0.9114	0.9114	0.9114	0.9114	0.9114	0.9114	0.9114

Table 9: COCO brown-rainy pose repair results. Metric: COCO keypoint AP.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	**Ours**
0.1	0.6048	0.6110	0.6132	0.5877	0.5985	0.6013	0.5950	**0.6183**
0.3	0.6040	0.6243	0.6215	0.6256	0.6153	0.6079	0.6165	**0.6306**
0.5	0.6209	0.6230	0.6282	0.6167	0.6206	0.6153	0.6182	**0.6317**
0.7	0.6151	0.6299	0.6293	0.6275	0.6256	0.6155	0.6223	**0.6319**
0.9	0.6091	0.6291	0.6170	0.6202	0.6213	0.6191	0.6301	**0.6355**
1.0	0.6259	0.6259	0.6259	0.6259	0.6259	0.6259	0.6259	0.6259

D Attribute-Slice Heatmaps

Figure 2 reports the corresponding ratio-0.5 COCO pose-estimation slice analysis over the full annotated attribute vocabulary. The targeted repair values are clothes color brown and rainy weather. As in the ImageNet10 example, the largest gains appear on the targeted values, while the remaining attributes show smaller changes.

E Hyperparameter Details

Table 11 reports the AttriSelect hyperparameters used in the main experiments. The parameters are the rarity exponent α, frequency-penalty weight λ, and class-balancing weight w, matching the notation in Eq. 5.

The larger frequency-penalty weight used for KITTI is consistent with the relative size of the generated repair set. All four settings add 1k generated images, but KITTI has only 5,000 source training images, compared with 10,400 for ImageNet10, 20,000 for COCO, and 80,000 for CelebA. The generated images therefore form a larger fraction of the KITTI repair candidate pool, making a stronger penalty on already-supplied attribute values useful for preserving source-data complementarity. This dataset-dependent shift supports the design of the score: the frequency-penalty term adapts the selector to the amount of generated repair data relative to the available source pool.

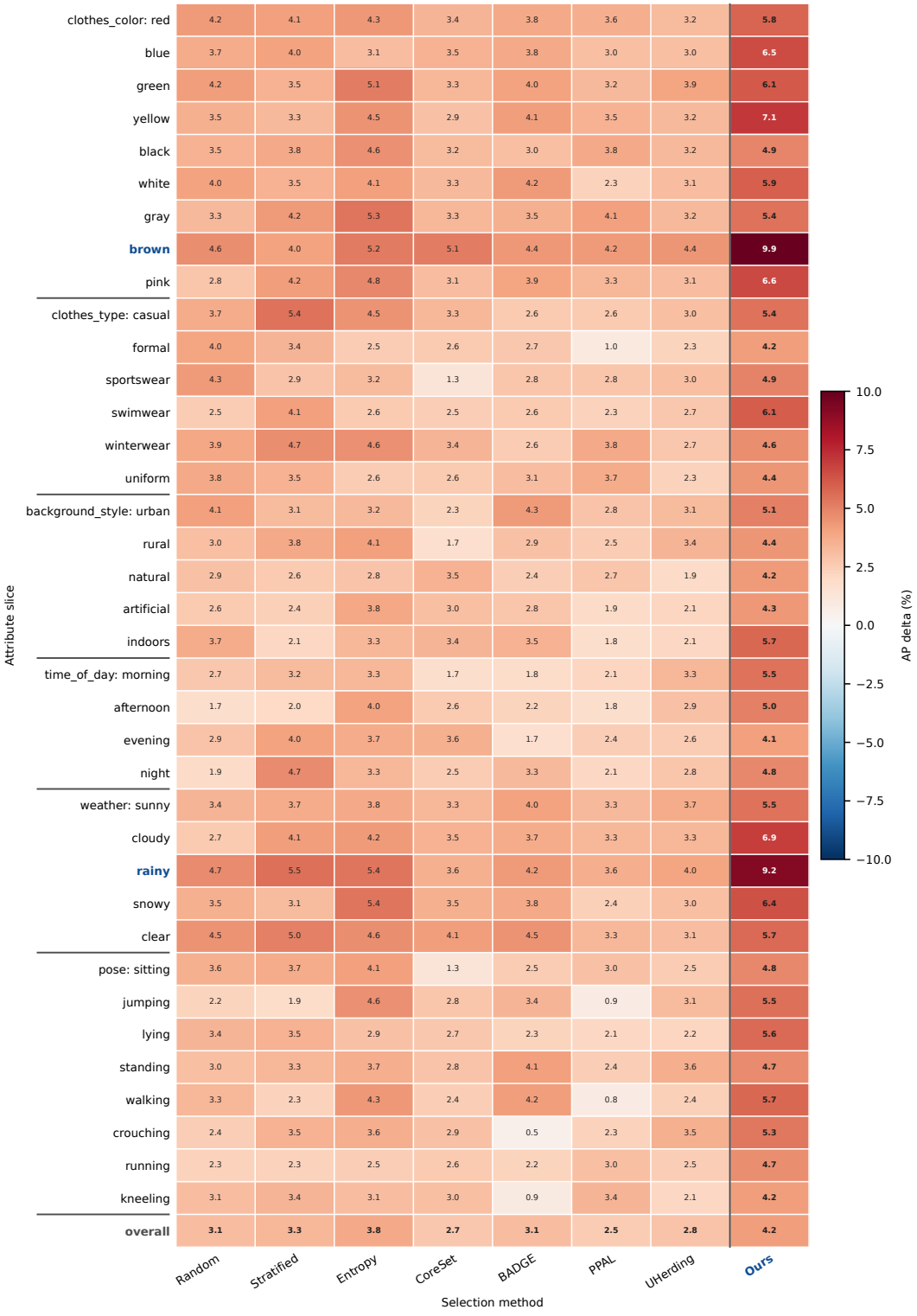


Figure 2: COCO brown-rainy ratio-0.5 attribute-slice AP deltas. The figure expands the slice-level analysis to all annotated COCO pose-estimation attribute groups. The targeted repair values are clothes color brown and rainy weather.

Table 10: KITTI red-cloudy object-detection repair results. Metric: test mAP50–95.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	**Ours**
0.1	0.5204	0.5308	0.5297	0.5237	0.5322	0.5222	0.5262	**0.5372**
0.3	0.5605	0.5572	0.5636	0.5612	0.5624	0.5573	0.5568	**0.5646**
0.5	0.5719	0.5680	0.5766	0.5759	0.5726	0.5675	0.5721	**0.5773**
0.7	0.5766	0.5760	0.5772	0.5772	0.5717	0.5768	**0.5782**	0.5780
0.9	0.5817	0.5830	0.5830	0.5781	0.5810	0.5838	0.5796	**0.5850**
1.0	0.5858	0.5858	0.5858	0.5858	0.5858	0.5858	0.5858	0.5858

Table 11: AttriSelect hyperparameters used in the main experiments.

Dataset	α	λ	w
ImageNet10	0.03	0.5	15
CelebA	0.03	0.5	15
COCO	0.03	0.5	15
KITTI	0.01	5.0	15

Table 12 gives the KITTI sensitivity across source-selection ratios. Across several AttriSelect parameter settings, the test mAP50–95 changes only modestly, suggesting that the selector is not brittle to small changes in α, λ, and w once the generated-data frequency penalty is present.

Table 12: KITTI AttriSelect sensitivity. Metric: test mAP50–95.

α	λ	w	0.1	0.3	0.5	0.7	0.9
0.01	5.0	15	**0.5372**	**0.5646**	**0.5773**	0.5780	**0.5850**
0.03	0.5	15	0.5358	0.5637	0.5764	0.5774	0.5842
0.03	5.0	8	0.5366	0.5641	0.5769	**0.5784**	0.5847
0.10	5.0	15	0.5349	0.5640	0.5768	0.5776	0.5839
0.80	10.0	4	0.5337	0.5629	0.5756	0.5769	0.5828

Table 13 shows the same pattern on COCO Brown-Rainy pose repair. The selected main setting remains strongest for most ratios, while nearby AttriSelect parameter settings stay within a narrow AP band.

F COCO Brown-Rainy Attribute-Score Component Ablation

Table 14 reports the corresponding component ablation for COCO Brown-Rainy pose repair. The full $R + P + C$ setting is strongest at four of the five ratios and remains close to the best result at the remaining ratio, suggesting that the three terms are complementary in this setting as well.

The leave-one-component-out comparisons show that removing any term reduces average AP relative to the full score, with average drops ranging from 0.00708 to 0.01428. Single-component variants are also less reliable than the full score across ratios. Overall, the COCO ablation supports the same conclusion as ImageNet10: the best repair behavior comes from combining rarity, generated-data frequency control, and class balancing rather than relying on any one term alone.

G Selection t-SNE Visualizations

Before measuring selected-set overlap directly, we first visualize the selected subsets in a shared low-dimensional embedding space. These plots are qualitative: the t-SNE coordinates are computed from lightweight image proxy embeddings and are used only for visualization, not for the selection-overlap metric below. Gray points denote source candidates, and colored points denote the subset selected by each method.

The ratio-0.10 plots in Figures 3–6 cover all four evaluated datasets. At this low selection ratio, each method selects a relatively small subset, so differences in coverage are visually readable. The

Table 13: COCO AttriSelect sensitivity. Metric: COCO keypoint AP.

α	λ	w	0.1	0.3	0.5	0.7	0.9
0.03	0.5	15	**0.6183**	**0.6306**	**0.6317**	0.6319	**0.6355**
0.01	0.5	15	0.6152	0.6298	0.6309	0.6308	0.6337
0.03	0.5	8	0.6165	0.6289	0.6305	**0.6323**	0.6344
0.03	0.5	0	0.6148	0.6276	0.6292	0.6306	0.6328
0.00	0.0	15	0.6042	0.6288	0.6203	0.6313	0.6268

Table 14: COCO Brown-Rainy attribute-score component ablation. Metric: COCO keypoint AP.

Ratio	none	R	P	C	$P+C$	$R+C$	$R+P$	$R+P+C$
0.1	0.6048	0.6099	0.5983	0.6067	0.6066	0.6149	0.5954	**0.6183**
0.3	0.6040	0.6134	0.6038	0.6136	0.6142	0.6196	0.6136	**0.6306**
0.5	0.6209	0.6074	0.6113	0.6149	0.6295	0.6246	0.6155	**0.6317**
0.7	0.6151	0.6222	0.6231	**0.6319**	0.6317	0.6302	0.6282	0.6311
0.9	0.6091	0.6175	0.6174	0.6270	0.6276	0.6225	0.6231	**0.6355**

available ratio-0.05 plots in Figures 8 and 9 provide a more stringent small-budget view for the datasets where this auxiliary diagnostic was run. CelebA and COCO are omitted at ratio 0.05 because the corresponding small-budget diagnostic was not included for those datasets. Figures 7 and 10 show compact KITTI overlays for a subset of methods, making the separation between AttriSelect and representative active-learning baselines easier to inspect.

The visual pattern should be interpreted in light of the selection score in Section 4.3. AttriSelect is not designed to be a uniformly spread coreset in the t-SNE plane. The rarity term $R_{i,a}$ intentionally gives higher priority to source examples whose attribute values are rare within their class, so the selected points can concentrate in regions that contain rare repair-relevant appearances. The penalty term $P_{i,a}$ reduces the priority of attribute values already supplied by the generated pool, so the selected source points are encouraged to complement the synthetic repair data instead of repeating it. Finally, the class-balance correction $C_c^{(k)}$ pulls the greedy selection away from a single high-scoring class or local region. Thus the expected signature is controlled non-uniformity: AttriSelect may put more mass on visually or semantically unusual regions than random, uncertainty, or pure coverage baselines, while still retaining cross-class coverage through the balance term.

The overlay plots illustrate this qualitative behavior on KITTI. The AttriSelect points do not simply trace the same support as the representative active-learning baselines; they are shifted toward a different subset of the proxy embedding space. This is consistent with the score's purpose: select source images that are representative enough to support retraining, but biased toward rare and repair-relevant attribute configurations. Because t-SNE is only a two-dimensional proxy visualization, this observation is not a standalone performance claim. It motivates the direct selected-set overlap analysis in Section H, where the same conclusion is tested through Jaccard overlap between selected subsets.

These visualizations should not be interpreted as proof that low overlap improves downstream repair. They instead provide a sanity check that AttriSelect is not visibly collapsing onto the same region as the standard active-learning selectors. The quantitative version of this diagnostic is the Jaccard analysis in Section H.

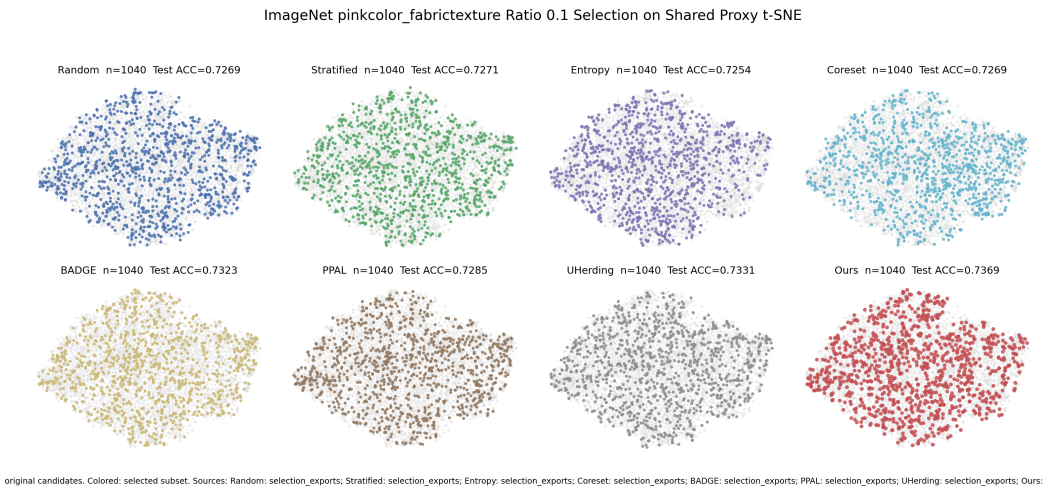


Figure 3: ImageNet ratio-0.10 selected subsets visualized on a shared source-image t-SNE embedding. Gray points show source candidates; colored points show the subset selected by each method.

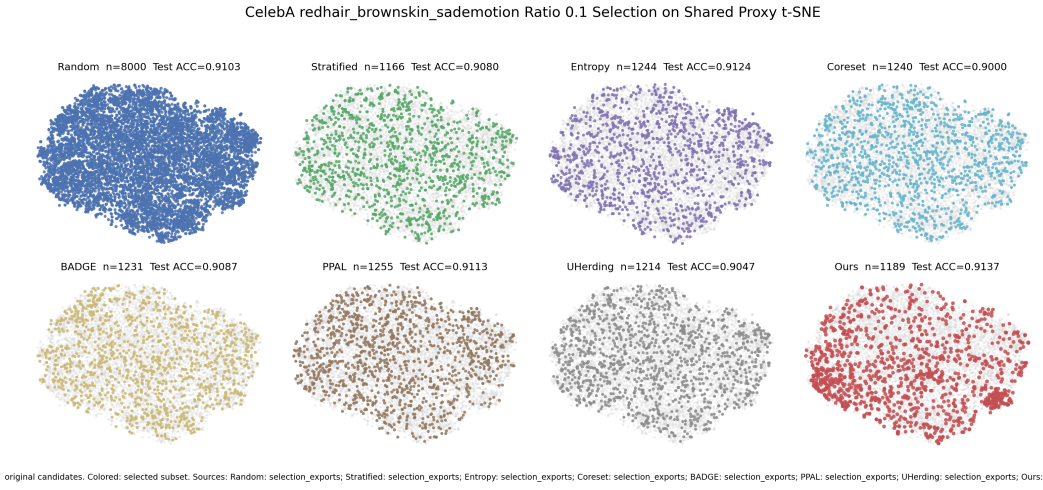


Figure 4: CelebA ratio-0.10 selected subsets visualized on a shared source-image t-SNE embedding.

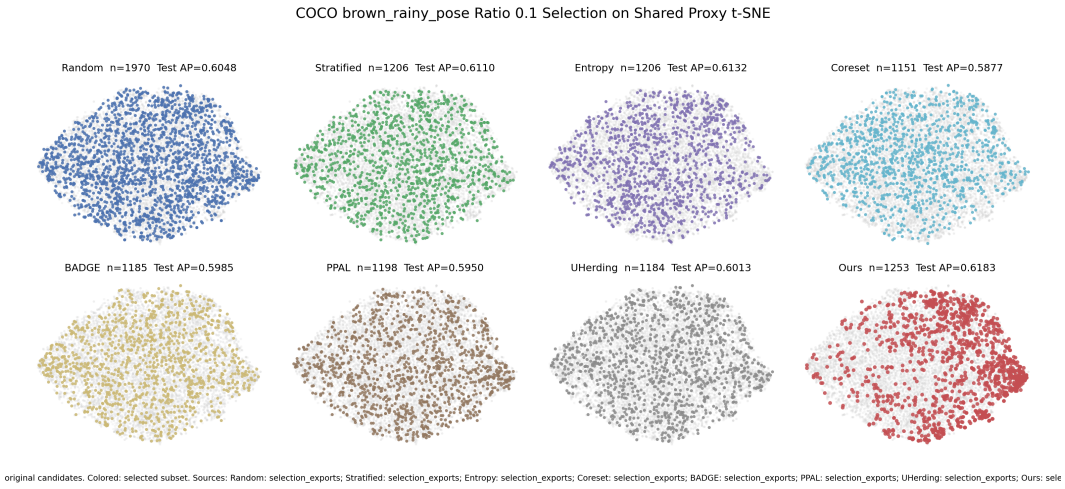


Figure 5: COCO Brown-Rainy ratio-0.10 selected subsets visualized on a shared source-image t-SNE embedding.

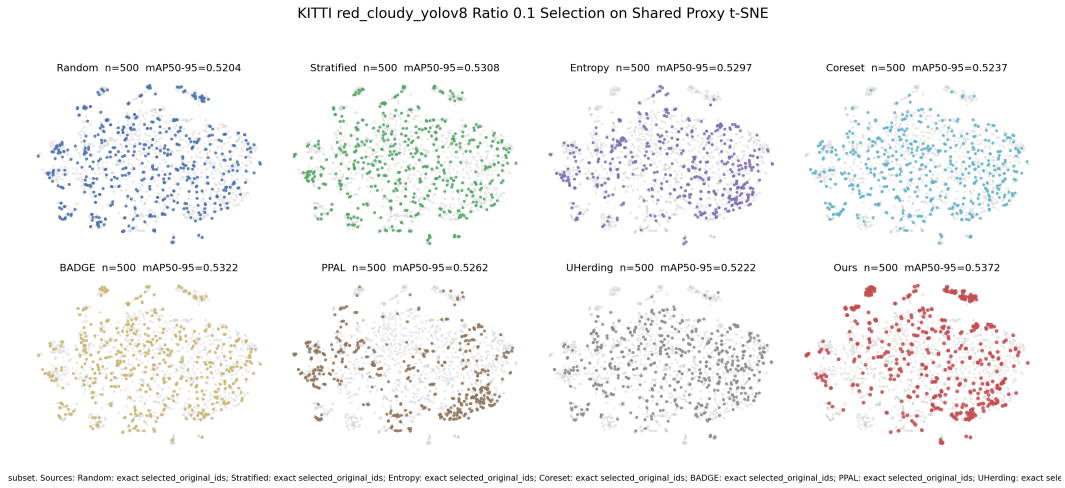


Figure 6: KITTI Red-Cloudy ratio-0.10 selected subsets visualized on a shared source-image t-SNE embedding.

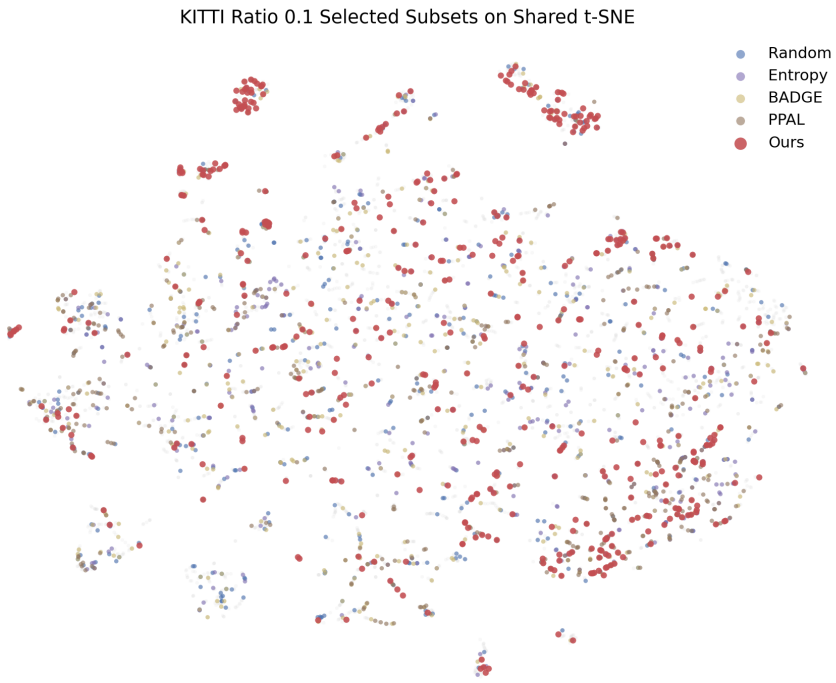


Figure 7: KITTI Red-Cloudy ratio-0.10 overlay view for selected active-learning baselines and AttriSelect.

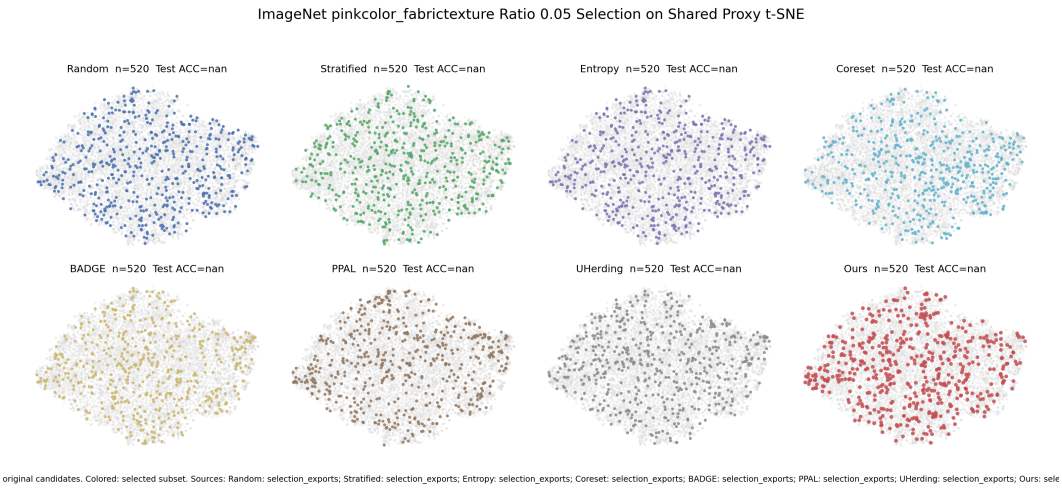


Figure 8: ImageNet ratio-0.05 selected subsets visualized on a shared source-image t-SNE embedding.

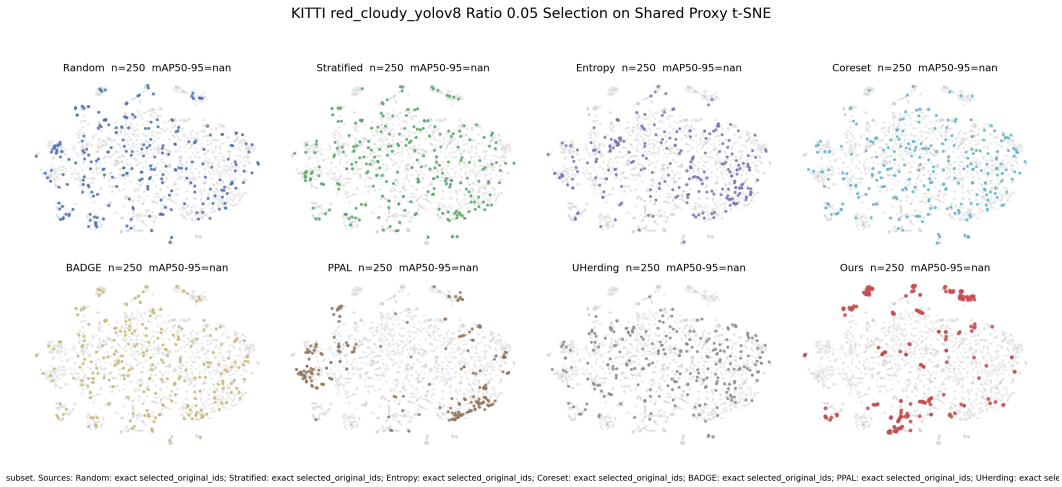


Figure 9: KITTI Red-Cloudy ratio-0.05 selected subsets visualized on a shared source-image t-SNE embedding.

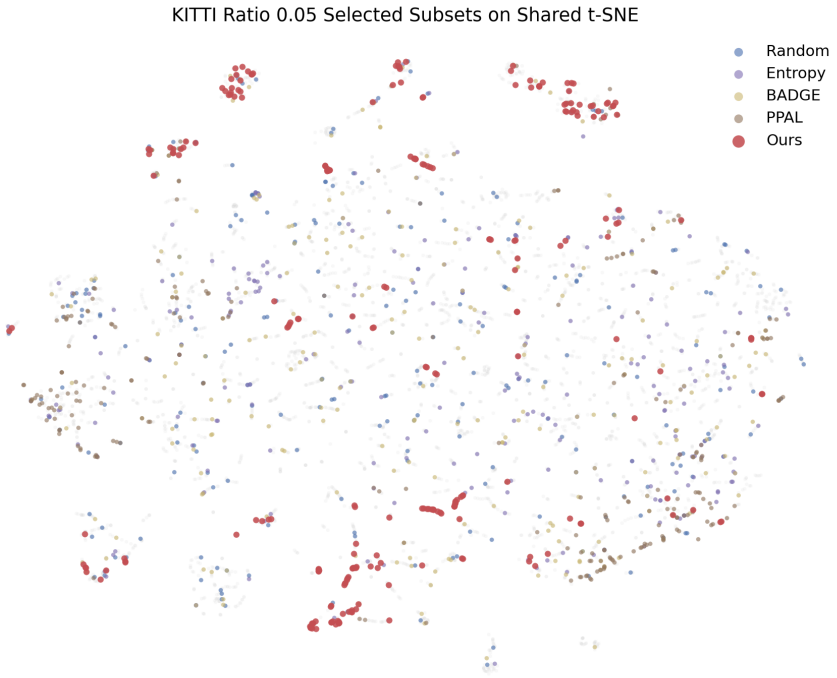


Figure 10: KITTI Red-Cloudy ratio-0.05 overlay view for selected active-learning baselines and AttriSelect.

H Selection Overlap as an Attribute-Information Diagnostic

This section formalizes what the selection-overlap plots test. Standard active learning selectors rank or cover examples using information such as logits, losses, uncertainty scores, feature embeddings, gradient embeddings, image distances, and sometimes class labels. Let H_i denote all such non-attribute information for source example x_i. Our selector additionally observes structured attribute metadata

$$A_i = \big(a(x_i) : a \in \mathcal{A}\big). \tag{29}$$

Thus the comparison is not only between two scoring formulas. It is a comparison between selectors that are measurable with respect to different information fields: generic AL is attribute-agnostic when it is a function of H_i, while the proposed selector is a function of both H_i and A_i through rarity, generated-data frequency, and class-aware re-ranking.

The overlap plots therefore answer a specific question: is AttriSelect merely reproducing the same subset chosen by uncertainty, CoreSet, BADGE, PPAL, Uncertainty Herding, or random selection? Low overlap by itself is not a performance guarantee. However, when low overlap is paired with the slice-enrichment result in Proposition 2 and with improved downstream repair, it supports a stronger interpretation: the gain comes from a semantic selection direction that is not already captured by generic AL signals.

Proposition 9 (Random-overlap scale for attribute-agnostic selectors)**.** *Let the source pool be partitioned into groups $\mathcal{G}_1, \dots, \mathcal{G}_K$, such as classes or class-balance bins, with $N_g = |\mathcal{G}_g|$. Let S_A be an attribute-driven selected set and let S_H be an attribute-agnostic selected set. Suppose S_A selects $m_{A,g}$ examples from group g and S_H selects $m_{H,g}$ examples from group g. If, conditional on the group counts and on S_A, the attribute-agnostic selector is exchangeable with respect to membership in S_A inside each group, then*

$$\mathbb{E}\big[|S_A \cap S_H| \mid S_A\big] = \sum_{g=1}^{K} \frac{m_{A,g} m_{H,g}}{N_g}. \tag{30}$$

In the ungrouped case, if both selectors choose $m = rN$ examples from a pool of size N, then

$$\mathbb{E}\big[|S_A \cap S_H| \mid S_A\big] = r^2 N. \tag{31}$$

Moreover, as $N \to \infty$ with fixed ratio r, the Jaccard overlap

$$J(S_A, S_H) = \frac{|S_A \cap S_H|}{|S_A \cup S_H|} \tag{32}$$

converges in probability to the random-subset scale

$$J_{\mathrm{rand}}(r) = \frac{r}{2 - r}. \tag{33}$$

Proof. Fix a group $\mathcal{G}_g$ and condition on S_A. Since S_H is exchangeable with respect to membership in S_A within the group, selecting $m_{H,g}$ examples from $\mathcal{G}_g$ gives a hypergeometric intersection with mean

$$m_{H,g} \frac{m_{A,g}}{N_g}. \tag{34}$$

Summing over groups gives the first claim. In the ungrouped case this reduces to $m^2/N = r^2 N$. Hypergeometric concentration gives $|S_A \cap S_H|/N \to r^2$ in probability. Since $|S_A \cup S_H| = 2rN - |S_A \cap S_H|$, the continuous mapping theorem gives

$$\frac{|S_A \cap S_H|}{|S_A \cup S_H|} \to \frac{r^2}{2r - r^2} = \frac{r}{2 - r}. \tag{35}$$

□

Proposition 9 gives a useful diagnostic. If an attribute-aware selector and an attribute-agnostic AL selector have Jaccard overlap close to $J_{\mathrm{rand}}(r)$, then the two selected sets are no more similar than two independent budget-matched subsets at that ratio. This is exactly what one expects when the attributes carry selection information not present in the AL variables H_i. Conversely, if a conventional

AL method were already aligned with the discovered attribute slice, its overlap with the selected set from AttriSelect would tend to rise above this random-overlap scale.

The ungrouped value $J_{\mathrm{rand}}(r)$ is a useful lower reference for selectors that share a group allocation over imbalanced groups. Suppose two exchangeable selectors both choose B_g examples from group g, with $\sum_g B_g = rN$ and $\sum_g N_g = N$. Proposition 9 gives expected intersection $\sum_g B_g^2/N_g$, and Cauchy–Schwarz gives

$$\sum_g \frac{B_g^2}{N_g} \geq \frac{\left(\sum_g B_g\right)^2}{\sum_g N_g} = r^2 N, \tag{36}$$

with equality only when B_g/N_g is constant across groups. For fixed total budget, the large-pool Jaccard reference is an increasing function of the intersection size, so class-balanced allocations on imbalanced datasets can have a random-overlap scale slightly above the ungrouped limit. This helps explain why the baseline–baseline averages in Table 15 are consistently above $J_{\mathrm{rand}}(r)$ when the baselines share similar class-balancing effects. When two methods use different group allocations, the full grouped expression in Proposition 9 is the appropriate reference.

Proposition 10 (Non-redundant repair utility)**.** *Assume the repair-utility model from Appendix B,*

$$U(S) = u_0|S| + (u_T - u_0)|S \cap T|, \qquad u_T > u_0, \tag{37}$$

where T is the target failure slice. Suppose AttriSelect satisfies the deterministic rare-slice enrichment condition in Proposition 3, while an attribute-agnostic selector satisfies the conditional-independence condition in Proposition 1, under the same per-class allocations B_c. Then the expected utility advantage of AttriSelect over the attribute-agnostic selector is at least

$$(u_T - u_0) \sum_c \left[\min\{B_c, |T_c|\} - B_c \frac{|T_c|}{N_c} \right]. \tag{38}$$

This lower bound is strict whenever at least one class has $0 < |T_c| < N_c$ and receives a positive budget that does not occupy the whole class, i.e., $0 < B_c < N_c$. If, in addition, the empirical Jaccard overlap between the two selected sets is close to $J_{\mathrm{rand}}(r)$, then this utility advantage cannot be explained by selecting the same source examples as the attribute-agnostic baseline.

Proof. Proposition 3 gives at least $\min\{B_c, |T_c|\}$ target examples for AttriSelect in class c. Proposition 1 gives $B_c|T_c|/N_c$ target examples in expectation for an attribute-agnostic selector under the same class allocation. Since both selectors use the same total budget, the $u_0|S|$ term cancels. Multiplying the difference in target-slice support by the marginal utility gap $u_T - u_0$ gives the stated bound. The strictness condition follows because $\min\{B_c, |T_c|\} > B_c|T_c|/N_c$ whenever a nontrivial target slice receives budget without occupying the whole class. The final statement follows from Proposition 9: near-random Jaccard overlap means the two selectors are not choosing substantially the same subset beyond what the sampling ratio itself forces. □

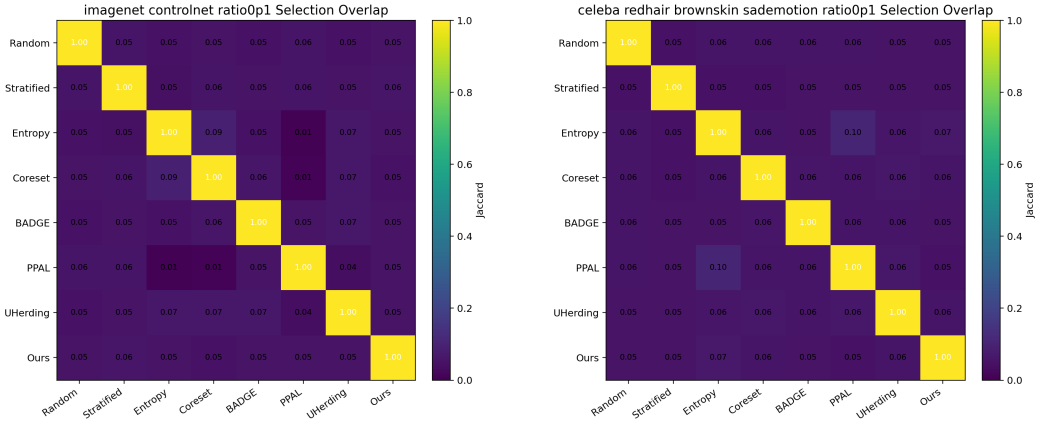
Table 15 summarizes the observed Jaccard overlaps between selected subsets. The t-SNE coordinates are not used in the calculation; the metric is computed only from selected-set membership. Across ratios, the average overlap between AttriSelect and the baselines is close to the random-subset scale $J_{\mathrm{rand}}(r)$ and lower than the average pairwise overlap among baselines. This supports the interpretation that AttriSelect operates along a different semantic information direction rather than imitating a conventional AL heuristic.

Table 15: Selection-overlap diagnostic. "Ours–baseline" is the mean Jaccard overlap between AttriSelect and the non-attribute baselines. "Baseline–baseline" is the mean pairwise Jaccard overlap among non-attribute baselines. Each row averages over ImageNet, CelebA, COCO, and KITTI.

Selection ratio	$J_{\mathrm{rand}}(r)$	Datasets	Ours–baseline	Baseline–baseline
0.10	0.0526	4	0.0533	0.0554
0.30	0.1765	4	0.1766	0.1828
0.50	0.3333	4	0.3323	0.3416

The corresponding heatmaps are shown in Figures 11, 12, and 13. The diagonal entries are one by definition; off-diagonal entries measure pairwise Jaccard overlap between selected source subsets.

These numbers and figures should be read together with the propositions above. They do not prove that lower overlap alone improves a model. Rather, they show that AttriSelect is not redundant with uncertainty, diversity, geometry, detector-difficulty, or gradient-embedding selectors. The improvement mechanism is therefore consistent with the theory: the selector spends budget on an attribute-defined repair signal that the generic AL variables do not explicitly encode.



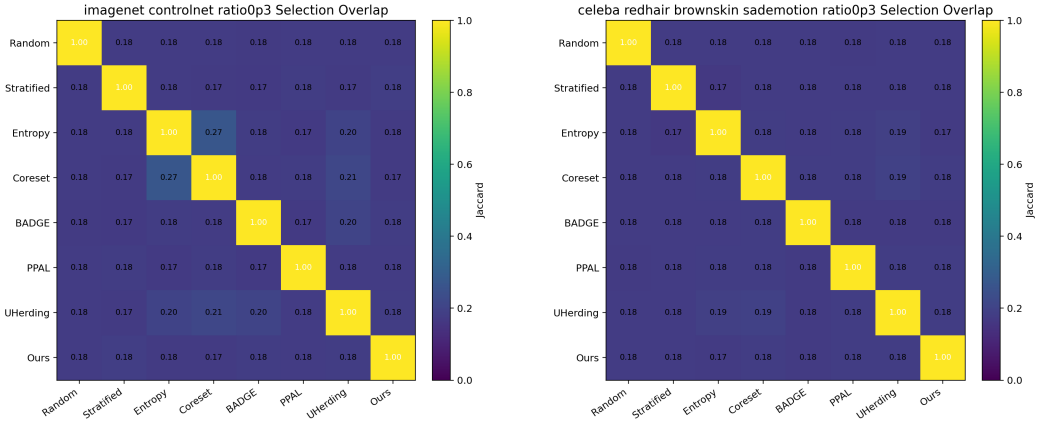
(a) ImageNet

(b) CelebA

(c) COCO

(d) KITTI

Figure 11: Pairwise Jaccard overlap among ratio 0.10 selected subsets.



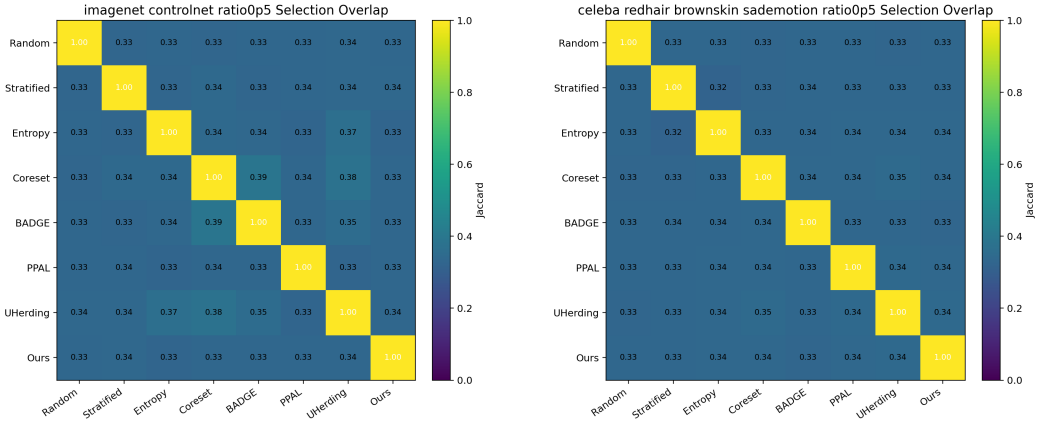
(a) ImageNet

(b) CelebA

(c) COCO

(d) KITTI

Figure 12: Pairwise Jaccard overlap among ratio 0.30 selected subsets.



(a) ImageNet

(b) CelebA

(c) COCO

(d) KITTI

Figure 13: Pairwise Jaccard overlap among ratio 0.50 selected subsets.